

DDA5001 Machine Learning

Overfitting and Validation

Yuning You

School of Science and Engineering
The Chinese University of Hong Kong, Shenzhen



Recap: Gradient Descent with Momentum

GD:

$$\theta_{k+1} = \theta_k - \mu_k \nabla \mathcal{L}(\theta_k)$$

A quite popular technique for accelerating gradient descent method is the **momentum** technique:

$$\theta_{k+1} = \theta_k - \mu_k \nabla \mathcal{L}(\theta_k) + \beta_k \underbrace{(\theta_k - \theta_{k-1})}_{\text{momentum}}$$

An equivalent form:

$$\begin{aligned}\theta_{k+1} &= \theta_k - m_k \\ m_k &= \mu_k \nabla \mathcal{L}(\theta_k) + \beta_k m_{k-1}\end{aligned}$$

- ▶ Each iteration takes nearly the same time cost as GD.
- ▶ Widely used in practice, notably in **Adam** algorithm.

Recap: Nesterov's Acceleration

GD:

$$\theta_{k+1} = \theta_k - \mu_k \nabla \mathcal{L}(\theta_k)$$

Another very useful technique to accelerate GD is **Nesterov's accelerated gradient descent (AGD)**:

$$\begin{aligned}\theta_{k+1} &= w_k - \mu_k \nabla \mathcal{L}(w_k) \\ w_k &= \theta_k + \frac{k-1}{k+2}(\theta_k - \theta_{k-1})\end{aligned}$$

- ▶ Each iteration takes nearly the same time cost as GD.
- ▶ Widely used in practice.

Agenda

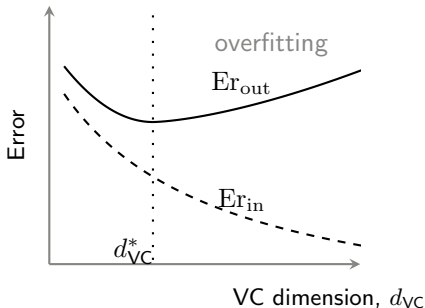
Overfitting

Validation

What is Overfitting?

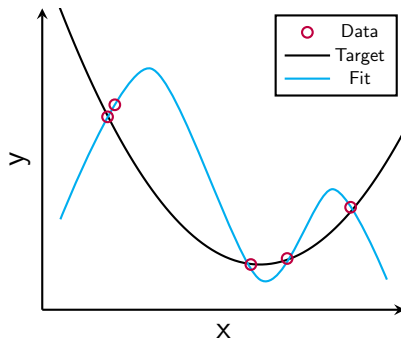
Overfitting

Fitting the data more than is needed



- ▶ One possible reason: $\mathcal{H}$ is more complex than is needed.
- ▶ It means fitting the observed data well no longer indicates that we will get a small out-of-sample error, and may actually lead to the opposite effect. That is, overfitting leads to small training error but bad generalization (large Er_{out}).

Overfitting: Example



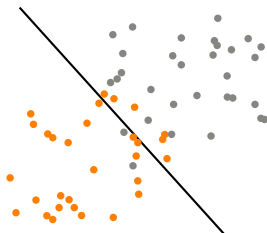
- ▶ Quadratic target function g .
- ▶ 5 data points with noise.
- ▶ 4-th order polynomial hypothesis set $\mathcal{H}$

$$Er_{in} = 0, \quad Er_{out} \text{ is huge}$$

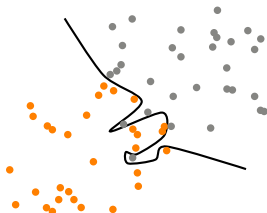
- ▶ The model uses its additional degrees of freedom to fit undesirable pattern in the data (for example, noise), yielding a final learned model that is inferior.

Overfitting: Example

Linear: nearly appropriate



Highly nonlinear: overfit



- ▶ We have linearly separable data plus noise, it becomes slightly non-linearly separable.
- ▶ Linear classifier

$Er_{in} \neq 0$, Er_{out} is appropriate

- ▶ **Highly nonlinear** hypothesis set $\mathcal{H}$

$Er_{in} = 0$, Er_{out} is larger

Overfitting: Catalysts

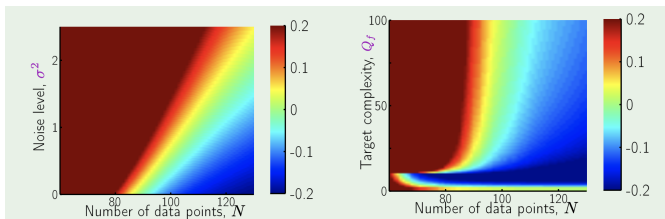


Figure: Color means the relative value of Er_{out} .

Overfitting: Catalysts (i.e., factors affecting overfitting)

- ▶ **Number of training samples** increase, overfitting decreases.
- ▶ **Noise** in data increase, overfitting increases.
- ▶ **Model complexity** increases, overfitting increases.

Overfitting: Catalysts

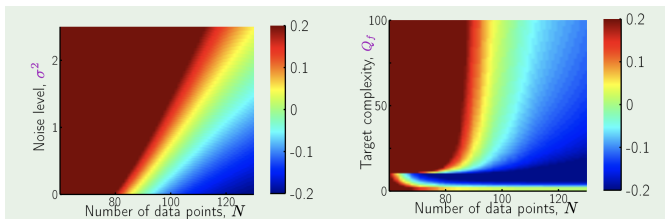


Figure: Color means the relative value of Er_{out} .

Overfitting: Catalysts (i.e., factors affecting overfitting)

- ▶ **Number of training samples** increase, overfitting decreases.
- ▶ **Noise** in data increase, overfitting increases.
- ▶ **Model complexity** increases, overfitting increases.

↪ Our HW2 will study the Catalysts for overfitting.

- ▶ One important approach to avoid overfitting is **validation**.
- ▶ Another one is **regularization**.

↪ We first study validation.

Agenda

Overfitting

Validation

Validation as A Cure for Overfitting

From VC generalization analysis, we have

$$\text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \text{overfit penalty}.$$

- ▶ Generalization error is large in the overfitting regime, hence it becomes the “overfit penalty”.
- ▶ The ideal case is to minimize Er_{out} directly, which is, however, not available.
- ▶ Can we turn to estimate the Er_{out} ?

Validation as A Cure for Overfitting

From VC generalization analysis, we have

$$\text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \text{overfit penalty.}$$

- ▶ Generalization error is large in the overfitting regime, hence it becomes the “overfit penalty”.
- ▶ The ideal case is to minimize Er_{out} directly, which is, however, not available.
- ▶ Can we turn to estimate the Er_{out} ?

Validation technique tries to estimate the out-of-sample error for eliminating overfitting:

$$\underbrace{\text{Er}_{\text{out}}(f)}_{\text{validation estimates this quantity}} \leq \text{Er}_{\text{in}}(f) + \text{overfit penalty.}$$

Where have we estimated Er_{out} ?

Validation as A Cure for Overfitting

From VC generalization analysis, we have

$$\text{Er}_{\text{out}}(f) \leq \text{Er}_{\text{in}}(f) + \text{overfit penalty.}$$

- ▶ Generalization error is large in the overfitting regime, hence it becomes the “overfit penalty”.
- ▶ The ideal case is to minimize Er_{out} directly, which is, however, not available.
- ▶ Can we turn to estimate the Er_{out} ?

Validation technique tries to estimate the out-of-sample error for eliminating overfitting:

$$\underbrace{\text{Er}_{\text{out}}(f)}_{\text{validation estimates this quantity}} \leq \text{Er}_{\text{in}}(f) + \text{overfit penalty.}$$

Where have we estimated Er_{out} ? Q5 in HW1. Why is it possible? Will justify later in the understanding of validation error.

Applications of Validation

Hyper-parameters: Something that are **NOT** automatically determined by the learning algorithm.

- ▶ Complexity of $\mathcal{H}$.
- ▶ Number of iterations in learning algorithm, i.e., stopping criterion.
- ▶ Learning rate.
- ▶ Regularization parameter λ (in our later lecture).
- ▶ ...

Applications of Validation

Hyper-parameters: Something that are **NOT** automatically determined by the learning algorithm.

- ▶ Complexity of $\mathcal{H}$.
- ▶ Number of iterations in learning algorithm, i.e., stopping criterion.
- ▶ Learning rate.
- ▶ Regularization parameter λ (in our later lecture).
- ▶ ...

In many cases, we have one (or more) hyper-parameters. The value chosen for hyper-parameters has a significant impact on the algorithm's output.

Model selection:

- ▶ The problem of selecting values for hyper-parameters is called **model selection**.
- ▶ **Validation** is used for model selection.

The Validation Dilemma

- ▶ What we have? **Training data**.
- ▶ Can we use the training data to estimate the Er_{out} and then select hyper-parameters (i.e., choose the hyper-parameter values which minimize the training error)?
- ▶ These hyper-parameters usually control the balance between **underfitting** and **overfitting**.
- ▶ Thus, it is not appropriate to let the **training data** influence the selection of hyper-parameters, as **this almost always leads to overfitting**.

Example: If we let the training data determine the degree in polynomial regression ($\mathcal{H}$ is the set of all polynomial functions with degree less than d), we will just end up choosing the maximum degree and doing interpolation (overfitting).

↪ We need some other data for estimating Er_{out} . In other words, we should not choose the hyper-parameters by minimizing the training error.

The Idea of Validation: Split Training Data

The idea

Split the training set to another 'training set' and a validation set.

Validation

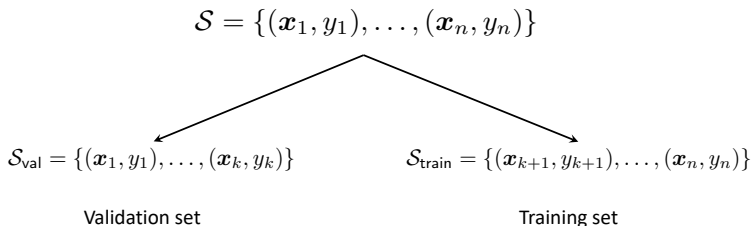
After learning $f_{\theta'} \in \mathcal{H}$ based on the new 'training set', use validation set to estimate $\text{Er}_{\text{out}}(f_{\theta'})$.

Why estimate $\text{Er}_{\text{out}}(f_{\theta'})$? We can see if the learned $f_{\theta'} \in \mathcal{H}$ is good or not in terms of (the estimated) out-of-sample error.

Recall the goal of machine learning is to make the out-of-sample error small. If $\text{Er}_{\text{out}}(f_{\theta'})$ is small, it means we are nearly done. If not, it means we have a bad learned model, either underfitting or overfitting.

The Validation Set

Split training data set:



- ▶ Data set: $\mathcal{S} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$, size n .
- ▶ Validation set: $\mathcal{S}_{\text{val}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_k, y_k)\}$, size k .
- ▶ New training set: $\mathcal{S}_{\text{train}} = \{(\mathbf{x}_{k+1}, y_{k+1}), \dots, (\mathbf{x}_n, y_n)\}$, size $n - k$.

Validation Error

Use the validation set to form an estimate (we use f' to represent $f_{\theta'}$ to ease notation)

$$\text{Er}_{\text{val}}(f') = \frac{1}{k} \sum_{i \in \mathcal{S}_{\text{val}}} e(f'(\mathbf{x}_i), y_i)$$

Validation Error

Use the validation set to form an estimate (we use f' to represent $f_{\theta'}$ to ease notation)

$$\text{Er}_{\text{val}}(f') = \frac{1}{k} \sum_{i \in \mathcal{S}_{\text{val}}} e(f'(\mathbf{x}_i), y_i)$$

How well $\text{Er}_{\text{val}}(f')$ approximates $\text{Er}_{\text{out}}(f')$?

► In expectation

$$\mathbb{E} [\text{Er}_{\text{val}}(f')] = \frac{1}{k} \sum_{i \in \mathcal{S}_{\text{val}}} \mathbb{E} [e(f'(\mathbf{x}_i), y_i)] = \text{Er}_{\text{out}}(f')$$

This implies that the validation error is an **unbiased** estimation of the out-of-sample error of f' .

However, unbiased estimation is too weak a guarantee. Can we have more?

Validation Error Approximates Out-of-Sample Error

Mimicking the derivation of the **generalization result for a single hypothesis**, we can utilize the Hoeffding's inequality to show that

$$\text{Er}_{\text{out}}(f') \leq \text{Er}_{\text{val}}(f') + \mathcal{O}\left(\frac{1}{\sqrt{k}}\right).$$

- The key of this result is that we only need to derive the bound for **a single hypothesis f'** , thus the derivation is exactly the same as the fixed f generalization result. No union bounds needed.

Validation Error Approximates Out-of-Sample Error

Mimicking the derivation of the **generalization result for a single hypothesis**, we can utilize the Hoeffding's inequality to show that

$$\text{Er}_{\text{out}}(f') \leq \text{Er}_{\text{val}}(f') + \mathcal{O}\left(\frac{1}{\sqrt{k}}\right).$$

- ▶ The key of this result is that we only need to derive the bound for **a single hypothesis f'** , thus the derivation is exactly the same as the fixed f generalization result. No union bounds needed.
- ▶ One key feature is that **$\text{Er}_{\text{val}}(f')$ is computable**.
- ▶ This means that once we have enough validation data, i.e., **k is large**, we get good estimation of $\text{Er}_{\text{out}}(f')$ by $\text{Er}_{\text{val}}(f')$. Then, we will know whether the model f' is good or not.

Validation Error Approximates Out-of-Sample Error

Mimicking the derivation of the **generalization result for a single hypothesis**, we can utilize the Hoeffding's inequality to show that

$$\text{Er}_{\text{out}}(f') \leq \text{Er}_{\text{val}}(f') + \mathcal{O}\left(\frac{1}{\sqrt{k}}\right).$$

- ▶ The key of this result is that we only need to derive the bound for **a single hypothesis f'** , thus the derivation is exactly the same as the fixed f generalization result. No union bounds needed.
- ▶ One key feature is that **$\text{Er}_{\text{val}}(f')$ is computable**.
- ▶ This means that once we have enough validation data, i.e., **k is large**, we get good estimation of $\text{Er}_{\text{out}}(f')$ by $\text{Er}_{\text{val}}(f')$. Then, we will know whether the model f' is good or not.
- ▶ This result applies to **test data** as well, i.e.,

$$\text{Er}_{\text{out}}(f') \leq \text{Er}_{\text{test}}(f') + \mathcal{O}(1/\sqrt{k})$$

if we have k test data points. The reasoning is exactly the same.

Any remaining issues?

Trade-off in Validation

- ▶ Where is k from? Where is f' from?

Trade-off in Validation

- Where is k from? Where is f' from?

$$\underbrace{n}_{\text{number of data}} = \underbrace{k}_{\text{number of validation data}} + \underbrace{n - k}_{\text{number of new training data}}$$

- $\text{Er}_{\text{val}}(f')$ is $\mathcal{O}(1/\sqrt{k})$ -close to $\text{Er}_{\text{out}}(f')$, while f' is learned by using $n - k$ new training data.
- Small k : Bad estimate of $\text{Er}_{\text{out}}(f')$.
- Large k : Good estimate of $\text{Er}_{\text{out}}(f')$, but bad f' (since few training samples).

Rule of thumb: $k = n/5$.

Restoring: k is Put Back to n

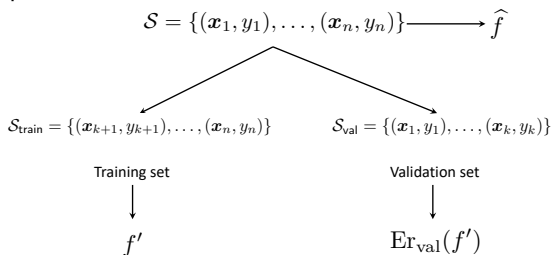
- Validation is to estimate $\text{Er}_{\text{out}}(f')$ using $\text{Er}_{\text{val}}(f')$, must we output f' as the learned model?

Restoring: k is Put Back to n

- ▶ Validation is to estimate $Er_{\text{out}}(f')$ using $Er_{\text{val}}(f')$, must we output f' as the learned model?
- ▶ No. It is mainly used to know how to choose **hyper-parameters** such as the best possible hypothesis and learning rate.

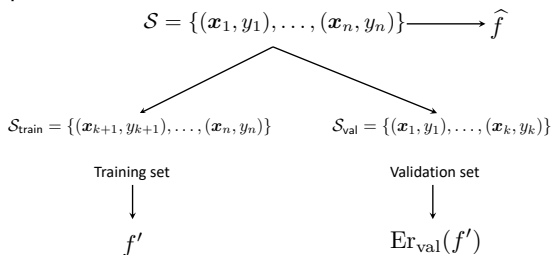
Restoring: k is Put Back to n

- ▶ Validation is to estimate $\text{Er}_{\text{out}}(f')$ using $\text{Er}_{\text{val}}(f')$, must we output f' as the learned model?
- ▶ No. It is mainly used to know how to choose **hyper-parameters** such as the best possible hypothesis and learning rate.
- ▶ Hence, after we have used the validation set to estimate the out-of-sample error, **re-train on the whole data set**.



Restoring: k is Put Back to n

- ▶ Validation is to estimate $\text{Er}_{\text{out}}(f')$ using $\text{Er}_{\text{val}}(f')$, must we output f' as the learned model?
- ▶ No. It is mainly used to know how to choose **hyper-parameters** such as the best possible hypothesis and learning rate.
- ▶ Hence, after we have used the validation set to estimate the out-of-sample error, **re-train on the whole data set**.



By using the previous bound, we have

$$\text{Er}_{\text{out}}(\hat{f}) \leq \text{Er}_{\text{out}}(f') \leq \text{Er}_{\text{val}}(f') + \mathcal{O}\left(\frac{1}{\sqrt{k}}\right).$$

The gray part is a **reasonable guess** (not proof) from VC analysis.

▪

Model Selection

Validation for Model Selection

We have discussed how to estimate $\text{Er}_{\text{out}}(f')$ using validation. It can be important to estimate how good our f' will perform on the test data.

- ▶ Indeed, validation can be used to **guide the learning process** systematically, by the way of **model selection**.
- ▶ Recall that model selection is to select **hyper-parameters**. These could be the choice between a linear model and a nonlinear model, the choice of order of polynomials in a model, the choice of learning rate, or any other choice that affects the learning process.

Validation for Model Selection

We have discussed how to estimate $\text{Er}_{\text{out}}(f')$ using validation. It can be important to estimate how good our f' will perform on the test data.

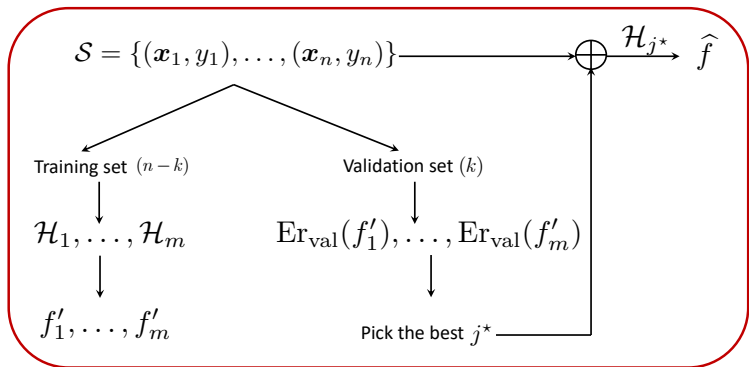
- ▶ Indeed, validation can be used to **guide the learning process** systematically, by the way of **model selection**.
- ▶ Recall that model selection is to select **hyper-parameters**. These could be the choice between a linear model and a nonlinear model, the choice of order of polynomials in a model, the choice of learning rate, or any other choice that affects the learning process.
- ▶ **Setup**: Suppose we have m candidate hypothesis spaces (can also be m different learning rate choices, etc)

$$\mathcal{H}_1, \dots, \mathcal{H}_m.$$

We can use the validation set to estimate the out-of-sample error by using $\text{Er}_{\text{val}}(f'_j)$ for each f'_j learned from these model spaces.

- ▶ **Selection**: Choose j^* such that $\text{Er}_{\text{val}}(f'_{j^*}) \leq \text{Er}_{\text{val}}(f'_j)$ for all j .
- ▶ **Restoring**: Train f on the whole set using model space $\mathcal{H}_{j^*}$, get $\hat{f}$.

Validation for Model Selection: illustration



Generalization Error of Model Selection

The model selection process gives a new hypothesis space consists of

$$\mathcal{H}_{\text{val}} = \{f'_1, f'_2, \dots, f'_m\}.$$

Model selection chooses **one** from $\mathcal{H}_{\text{val}}$ that achieves the smallest validation error.

- ▶ This process is equivalent to learning a model from $\mathcal{H}_{\text{val}}$ using the validation set, where Er_{val} is the “in-sample” error.
- ▶ This setting allows us to apply the generalization analysis for **finite hypotheses space** case, since $\mathcal{H}_{\text{val}}$ only consists of m hypotheses, i.e., $|\mathcal{H}_{\text{val}}| = m$.
- ▶ This gives

$$\text{Er}_{\text{out}}(\hat{f}) \leq \text{Er}_{\text{out}}(f'_{j^*}) \leq \text{Er}_{\text{val}}(f'_{j^*}) + \mathcal{O}\left(\sqrt{\frac{\log m}{k}}\right).$$

Again, the gray part is a reasonable guess from VC analysis.

Validation vs. Testing

- ▶ We call this “validation”, but how is it different from “testing”?
- ▶ Typically, validation is used to **make learning choices**, i.e., choosing hyper-parameters to avoid overfitting.

However,

The test data can never influence the training phase in any way.

If it impacts the learning process, i.e., which final $\hat{f} \in \mathcal{H}$ we choose, then **it is no longer a test set,**

it becomes a validation set.

Validation Dilemma

What we would like to have

$$\text{Er}_{\text{out}}(\hat{f}) \approx \text{Er}_{\text{out}}(f') \approx \text{Er}_{\text{val}}(f')$$

↑ ↑

small k large k

- All we need to do is set k so that it is **simultaneously** small and large...

Is it possible? Yes.

⇒ Next lecture: Cross validation and regularization.